EXP 26: Prolog Program for Fruit and its Color using Backtracking

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Aim:

To develop a Prolog program that stores fruits with their colors as facts, and demonstrates backtracking by finding all fruits that match a given color.

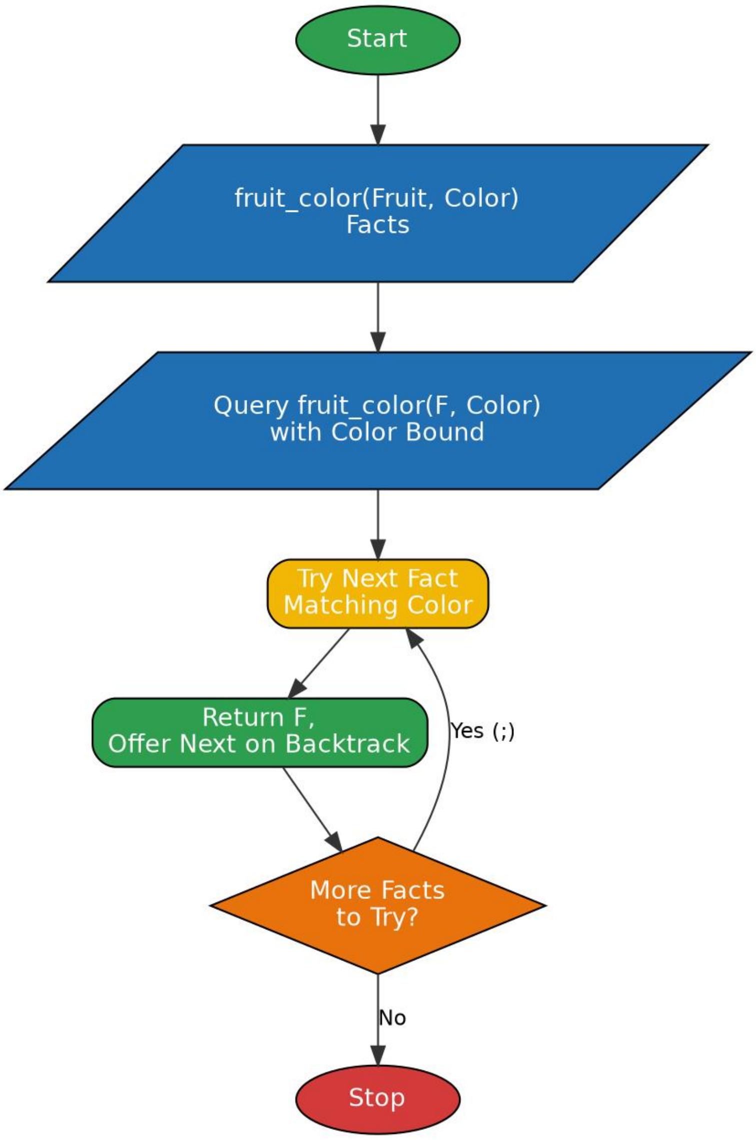
Algorithm:

Step 1: Start the program.
Step 2: Create a database containing fruits and their corresponding colors.
Step 3: Select a fruit or color for querying.
Step 4: Search the database for matching records.
Step 5: Display the color of the selected fruit or the fruits of the selected color.
Step 6: If multiple matches exist, use backtracking to find the next match.
Step 7: Continue until all matching records are displayed.
Step 8: Show the final results.
Step 9: Stop the program.

Source Code:

```
% Program for Fruit and its Color using Backtracking


fruit_color(apple, red).

fruit_color(banana, yellow).

fruit_color(grape, green).

fruit_color(orange, orange_color).

fruit_color(mango, yellow).

fruit_color(cherry, red).


show_color(Fruit) :-

   fruit_color(Fruit, Color),

   format("~w is ~w~n", [Fruit, Color]).


show_fruits(Color) :-

   forall(fruit_color(Fruit, Color),

        format("~w~n", [Fruit])).
```

Flowchart:


Start
fruit_color(Fruit, Color)
Facts
Query fruit_color(F, Color)
with Color Bound
Try Next Fact
Matching Color
Return F,
Offer Next on Backtrack
Yes (;)
More Facts
to Try?
No
Stop

Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [fruits].
true.

?- show_color(apple).
apple is red
true.

?- show_color(mango).
mango is yellow
true.

?- show_fruits(yellow).
banana
mango
true.

?- show_fruits(red).
apple
cherry
true.
```

Result:

Thus the Prolog program for fruit and its color using backtracking was written, executed successfully and the output was verified.